

Twelve-Year Arc of Code World Models (2014-2026)

Neural execution

Learning to Execute (1410.4615)

Neural Programmer-Interpreters

Dynamic Neural Program Embedding

Neural Code Fusion

IPA-GNN (interpreter line stalls)

World models / RL

Ha & Schmidhuber

RAP (LLM-as-WM + MCTS)

Generating CWMs via MCTS

CoLA

WebBeamer

CWM

Debugging CWMs / Reinforcement WM Learning

Code LLMs

Codex / Show Your Work

CodeRL

CodeExecutor / TRACED

CodeAct / SWE-agent

RLEF

SWE-RL

DeepSeek-R1

ATLAS / CLEVER (verifier-grounded)

Industrial CWM, ARC executable WMs

◆ marks first paper to use "world model" / "Dreamer" in the contribution name. ◆◆ CWM = named open-weights artifact.